

Industrial Automation module 2023-2024

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HomeWork

student:

The motion of the two axes (X,Y) of a 2D laser cutting machine is driven by two identical DC motors. The rotational motion of the motors is transformed into translation by means of an appropriate mechanism.

The parameters of the motor are the following:

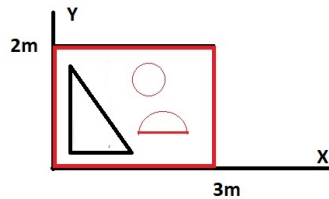
- $R_a = 1\Omega$; $L_a = 1.0\text{ mH}$
- $J = 0.12$; $B = 0.03$ (motor + load)
- $K_e = 0.6 \frac{\text{Volts} \cdot \text{sec}}{\text{rad}}$; $K_t = 0.6 \frac{\text{N} \cdot \text{m}}{\text{A}}$
- Max power = 4KW; Max Armature current = 30A; Max Voltage = 200V

The mechanism that transforms the rotation into translation has a transformation coefficient equal to: $1\text{ m} = 800\text{ rev.}$

Moreover for each motor it is possible to measure the armature current, the rotation speed ($\frac{\text{rad}}{\text{sec}}$) and the angle (rad).

The student has to develop:

1. A reference signals such that the cutting machine is able to extract a circle, a triangle and a semicircle from a rectangular slab (3m long and 2m high). The size, placement and orientation of the figures are left to the student. The rest position of the machine is in upper right corner of the plate.
2. The feedback controller for each motor that guarantees a maximum position error of less than 2 mm even when there is an uncertainty of parameters J and B of 10% with respect to the nominal ones.



the student will also have to develop a finite state machine and its possible implementation (Ladder and or SFC) for machine management.